

RAZI ASHARY

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A geodetic graduate turned AI researcher and developer, driven by a passion for learning, collaboration, and innovation. Skilled in architecting agentic AI systems, memory and retrieval infrastructure, and production-grade multi-agent platforms. Currently exploring the limitless possibilities of AI to create meaningful impacts across industries.

Experience

Head of Research - Indonesia

May 2025 – Present

AIRS Foundry

- **AIRS Forge (Cognition Engine):** Architected AIRS Forge, the company's core intelligence layer — a domain-agnostic “cognitive OS” that compiles any source (chat, documents, code, telemetry, media) into structured, governed, fully-cited machine memory across four cognitive pillars over one shared substrate, built on a single Postgres store (pgvector + Apache AGE) with content-addressed identity, queryable provenance, a visibility vault, and forced row-level multi-tenant isolation.
- **Auto-Storage (shipped):** Designed and shipped Auto-Storage, a deterministic, source-aware memory compiler with 12+ ingestion adapters and 14-language code parsing — proven on a live database at scale by compiling 100,000 code files and 20,000 document pages into ~340,000 structured atoms and ~240,000 typed relationships with zero errors.
- **Auto-Research (built):** Built Auto-Research, a retrieval and reasoning engine that returns cited evidence or an honest abstention instead of hallucinating — measured 0% hallucination, ~89–94% citation accuracy, and correct retrieval across all 14 source types; implemented as a deterministic 16-state pipeline with an inspectable 24-dimension ranking function, exposed identically over in-process, REST, gRPC, and MCP.
- **Core-Orchestrator & Multi-Agent Platform:** Designed and led the core-orchestrator, a Python/FastAPI multi-agent backend with 21 lifecycle state machines, Google Agent-to-Agent (A2A) protocol, native MCP tool-calling, SSE token streaming, and a multi-provider LLM layer — including a proprietary Adobe Compiler that cut per-call token consumption by ~75% for enterprise creative-production workflows.
- **AutoDev Platform:** Architected AutoDev, a model-agnostic, downloadable AI developer-agent platform (Tauri v2 / Rust / React / Python) that orchestrates a fleet of specialized agents (Supervisor, QC, Workers) on any LLM provider from a single binary — 1,200+ documented features spanning Skills, Recipes, Subagents, MCP servers, and Hooks, with zero vendor lock-in.
- **Memory Infrastructure:** Engineered a universal Memory MCP service with episodic, semantic, and procedural tiers over a hot/warm/cold storage architecture, kept provider-agnostic (Mem0, Redis HNSW, Neo4j graph memory) to avoid vendor lock-in and enable a phased migration strategy with full rollback.
- **Blueprinting & System Innovation:** Designed novel system architectures across product verticals, including an end-to-end speech-to-speech translation pipeline (LiveKit + Deepgram + Cartesia/ElevenLabs hybrid streaming), a node-based visual builder for chaining APIs and agent workflows, and a chemistry/periodic-table-inspired memory model — establishing the technical direction for AIRS product development.
- **Cross-Departmental Technical Guidance:** Serve as the primary technical reference across the company, providing architecture, system-design, and AI-integration direction to the manufacturing, marketing, and operations teams.
- **Cloud Infrastructure & Deployment:** Manage GCP infrastructure — Compute Engine VMs (Jakarta region), Cloud Run services, Docker Compose orchestration, S3-to-GCS bucket migrations, and cross-account deployments — ensuring stable, scalable multi-service operations for AIRS product environments.
- **Research & Knowledge Management:** Conduct weekly research on AI models, tools, frameworks, and papers, producing monthly findings documents with applicability notes for company integration and advancing selected findings into active prototypes or formal proposals.

Artificial Intelligence Research and Development - Indonesia

Nov 2023 – Mar 2025

Ai&You

- **Research & Innovation:** Conduct ongoing research on 1–2 AI models, tools, or frameworks weekly to stay at the forefront of AI advancements.
- **Blueprinting & Tech Architecture:** Designed 3 proprietary AI workflow architectures, establishing essential guidelines for tool development within the company.
- **Minimum Viable Product (MVP) Creation:** Delivered 8 MVPs, including a key AI workflow for the company and various client projects, driving successful project launches.
- **Data Engineering & Analytics:** Developed a comprehensive chatbot backend pipeline using Python for a financial consultant company, enhancing data analysis and decision-making processes.
- **Cognitive AI:** Used GPT APIs to instruct LLMs toward specific tasks and behaviors. Implemented Retrieval-Augmented Generation (RAG) for Visual Question Answering (VQA) to build an assistant that identifies specific car brands and

models, and researched LLM architecture, reasoning capabilities, and development techniques using LangChain and NVIDIA NIM.

- **Generative AI:** Fine-tuned and instructed 3 generative AI models (Stable Diffusion, EpicRealism, Juggernaut) for the beauty industry, contributing to innovative image and video applications.
- **Node-Based Chatbot Development:** Developed 5 high-performing chatbots averaging 100–200 daily users, including solutions for prominent clients like Mercedes.
- **Team Leadership:** Led a growing team of 2 developers, ensuring successful project execution and continuous improvement in AI implementations.
- **Feasibility Studies:** Conducted feasibility studies on all incoming projects, ensuring alignment with company objectives and client needs.
- **Client Engagement & Use Case Development:** Developed 10 use cases, including a successful deployment of a customer-facing chatbot for Mercedes at an automotive event.
- **Public Speaking & Teaching:** Presented at 2 industry events to audiences of 60+, educating on AI fundamentals and fostering greater AI adoption.

Research Staff — QZSS Satellite for Heading Determination Project - Bandung, Indonesia Jun 2023 – Jul 2023
Bandung Institute of Technology

- Acquired boat heading and positioning data using the QZSS satellite system to assess its accuracy for marine navigation.
- Conducted bathymetric surveys with a Multi-beam Echosounder, supported by Sound Velocity Profiler and Side Scan Sonar data, to enhance underwater mapping precision.
- Analyzed and visualized boat heading accuracy using MATLAB, comparing QZSS data with other satellite systems to evaluate performance improvements.

Assistant Lecturer — GNSS Survey Subject - Bandung, Indonesia Aug 2022 – Dec 2022
Bandung Institute of Technology

- Developed specialized modules on Real-Time Kinematic (RTK) surveying and GNSS data processing and analysis, providing hands-on learning experiences for students.
- Managed around 100 students, organizing practical sessions, supervising fieldwork, and guiding project development across 10 teams.
- Led a comprehensive final project involving an end-to-end GNSS field survey, overseeing data acquisition and ensuring accurate data processing by the student teams.

Research Staff — Green Energy Bawean Island Project - Bawean Island, Indonesia Jul 2022 – Oct 2022
Bandung Institute of Technology

- Measured potential hydropower and solar power generation for green-energy feasibility studies, using Automatic Weather Stations (AWS) to collect data on solar irradiance and other parameters such as humidity.
- Conducted data analysis to understand the correlation between hydropower generation and solar readings, contributing to research documentation and paper writing.
- Coordinated a collaborative research project with two universities, focusing on hydropower measurements and interdisciplinary data integration.

Education

Bandung Institute of Technology - Bandung, Indonesia Aug 2019 – Jul 2023
Bachelor Degree in Geodesy and Geomatics Engineering — GPA 3.54/4.00

- Cumlaude Graduate
- Dean's List in the sixth semester

SMAN 3 Bandung - Bandung, Indonesia Apr 2016 – May 2019
High School Diploma in Science and Applied Science

- Head of Sponsorship for a book event

Organisational Experience

Head of Profession Bureau - Bandung, Indonesia Jun 2022 – Apr 2023
Ikatan Mahasiswa Geodesi ITB (IMG-ITB)

- **Geodesy Survey Teaching:** Led a team conducting practical training on geodesy surveys, including aerial surveys using drones, bathymetric surveys with Single Beam Echo Sounders (SBES), and land surveys with Terrestrial Laser Scanners.

- **Team Management:** Oversaw two teams — one focused on the practical applications of geodesy and the other on research, producing three papers with one published in a journal.
- **Bureau Leadership:** Managed a bureau of around 300 members, organizing events, coordinating with organization leadership, handling administration, and making decisions on the professional aspects of the bureau.

Chief Committee of April Graduation - Bandung, Indonesia

Jan 2021 – Apr 2021

Ikatan Mahasiswa Geodesi ITB (IMG-ITB)

- **Event Supervision:** Oversaw all aspects of the IMG-ITB April graduation events, ensuring seamless execution and adherence to schedule and quality standards.
- **Leadership and Coordination:** Led a committee of approximately 50 members, delegating tasks, managing resources, and guiding the team to achieve event objectives effectively.
- **Innovation:** Created a new way of celebrating the geodesy student association's graduation tradition during COVID-19.
- **Decision Making:** Acted as the primary decision-maker for the committee on event logistics, coordination, and problem-solving to ensure a successful ceremony.

Capabilities

- **Programming Languages:** Python, TypeScript, Rust, HTML5, CSS, JavaScript, SQL, MATLAB
- **Tools, Frameworks & Libraries:** FastAPI, PyTorch, TensorFlow, scikit-learn, LangChain, Tauri, React, Node.js, PostgreSQL (pgvector) & Apache AGE, Redis, Qdrant, Docker, Docker Compose, MCP Server Development, Google A2A, LiteLLM, LiveKit, GCP (Compute Engine, Cloud Run, Vertex AI), YOLO, ComfyUI, GitHub, GitLab, Figma, Windsurf, Claude Code
- **Soft Skills:** Leadership, Teamwork, Communication, Problem Solving, Adaptability, Decision Making, Project Management
- **Languages:** English, Indonesian

Projects

AIRS Forge - AIRS Foundry

A domain-agnostic cognition and memory platform (a “cognitive OS layer”) that compiles any source into structured, governed, cited memory and reasons over it. Built as four physics-named pillars — Auto-Storage (chemistry), Auto-Research (physics), Auto-Learn (thermodynamics), and Auto-Execute (mechanics) — over one deterministic Postgres substrate, exposed identically over in-process, REST, gRPC, and MCP with a committed OpenAPI 3.1 contract. Auto-Storage is shipped and proven at scale (340K+ atoms, 0 errors); Auto-Research is built with 0% hallucination and ~89–94% citation accuracy.

AutoDev - AIRS Foundry

A model-agnostic, downloadable AI developer-agent platform built for enterprise-grade agent orchestration with zero vendor lock-in. Any AIRS team member or worker agent can download a single binary, plug in any LLM API key, and immediately access a full agentic capability stack including AUTODEV.md, Skills, Recipes, Subagents, MCP servers, and Hooks. Purpose-built for the AIRS Foundry AutoDev worker pipeline.